

Chapter 9 supplementary material

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Matrix expression of four-factor model for PCS items

$$\mathbf{Y} = \mathbf{\Lambda}\boldsymbol{\eta} + \boldsymbol{\varepsilon}$$

$$\begin{bmatrix} TE1 \\ TE2 \\ TE3 \\ TE4 \\ TE5 \\ OE1 \\ OE2 \\ OE3 \\ OE4 \\ LV1 \\ LV2 \\ LV3 \\ LV4 \\ EM1 \\ EM2 \\ EM3 \\ EM4 \\ EM5 \\ EM6 \end{bmatrix} = \begin{bmatrix} \lambda_{1,1} & 0 & 0 & 0 \\ \lambda_{2,1} & 0 & 0 & 0 \\ \lambda_{3,1} & 0 & 0 & 0 \\ \lambda_{4,1} & 0 & 0 & 0 \\ \lambda_{5,1} & 0 & 0 & 0 \\ 0 & \lambda_{6,2} & 0 & 0 \\ 0 & \lambda_{7,2} & 0 & 0 \\ 0 & \lambda_{8,2} & 0 & 0 \\ 0 & \lambda_{9,2} & 0 & 0 \\ 0 & 0 & \lambda_{10,3} & 0 \\ 0 & 0 & \lambda_{11,3} & 0 \\ 0 & 0 & \lambda_{12,3} & 0 \\ 0 & 0 & \lambda_{13,3} & 0 \\ 0 & 0 & 0 & \lambda_{14,4} \\ 0 & 0 & 0 & \lambda_{15,4} \\ 0 & 0 & 0 & \lambda_{16,4} \\ 0 & 0 & 0 & \lambda_{17,4} \\ 0 & 0 & 0 & \lambda_{18,4} \\ 0 & 0 & 0 & \lambda_{19,4} \end{bmatrix} \begin{bmatrix} TE \\ OE \\ LV \\ EM \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \\ \varepsilon_9 \\ \varepsilon_{10} \\ \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{15} \\ \varepsilon_{16} \\ \varepsilon_{17} \\ \varepsilon_{18} \\ \varepsilon_{19} \end{bmatrix}$$

Thus, the factor loading matrix is

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_{1,1} & 0 & 0 & 0 \\ \lambda_{2,1} & 0 & 0 & 0 \\ \lambda_{3,1} & 0 & 0 & 0 \\ \lambda_{4,1} & 0 & 0 & 0 \\ \lambda_{5,1} & 0 & 0 & 0 \\ 0 & \lambda_{6,2} & 0 & 0 \\ 0 & \lambda_{7,2} & 0 & 0 \\ 0 & \lambda_{8,2} & 0 & 0 \\ 0 & \lambda_{9,2} & 0 & 0 \\ 0 & 0 & \lambda_{10,3} & 0 \\ 0 & 0 & \lambda_{11,3} & 0 \\ 0 & 0 & \lambda_{12,3} & 0 \\ 0 & 0 & \lambda_{13,3} & 0 \\ 0 & 0 & 0 & \lambda_{14,4} \\ 0 & 0 & 0 & \lambda_{15,4} \\ 0 & 0 & 0 & \lambda_{16,4} \\ 0 & 0 & 0 & \lambda_{17,4} \\ 0 & 0 & 0 & \lambda_{18,4} \\ 0 & 0 & 0 & \lambda_{19,4} \end{bmatrix}$$

To scale the latent variables, the first non-zero factor loading in each column might be fixed to 1, such that

$$\mathbf{\Lambda} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \lambda_{2,1} & 0 & 0 & 0 \\ \lambda_{3,1} & 0 & 0 & 0 \\ \lambda_{4,1} & 0 & 0 & 0 \\ \lambda_{5,1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \lambda_{7,2} & 0 & 0 \\ 0 & \lambda_{8,2} & 0 & 0 \\ 0 & \lambda_{9,2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_{11,3} & 0 \\ 0 & 0 & \lambda_{12,3} & 0 \\ 0 & 0 & \lambda_{13,3} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \lambda_{15,4} \\ 0 & 0 & 0 & \lambda_{16,4} \\ 0 & 0 & 0 & \lambda_{17,4} \\ 0 & 0 & 0 & \lambda_{18,4} \\ 0 & 0 & 0 & \lambda_{19,4} \end{bmatrix}$$

The inter-factor covariance matrix is

$$\mathbf{\Psi} = \begin{bmatrix} \sigma_{\eta_1}^2 & & & \\ \sigma_{\eta_2\eta_1} & \sigma_{\eta_2}^2 & & \\ \sigma_{\eta_3\eta_1} & \sigma_{\eta_3\eta_2} & \sigma_{\eta_3}^2 & \\ \sigma_{\eta_4\eta_1} & \sigma_{\eta_4\eta_2} & \sigma_{\eta_4\eta_3} & \sigma_{\eta_4}^2 \end{bmatrix}$$

where $\sigma_{\eta_1}^2$ is the variance of η_1 (i.e., variance of the *TE* factor), $\sigma_{\eta_2}^2$ is the variance of η_2 (i.e., variance of *OE*), $\sigma_{\eta_3}^2$ is the variance of η_3 (i.e., variance of *LV*), and $\sigma_{\eta_4}^2$ is the variance of η_4 (i.e., variance of *EM*).

Next, $\sigma_{\eta_2\eta_1}$ is the covariance between η_1 and η_2 , $\sigma_{\eta_3\eta_1}$ is the covariance between η_1 and η_3 , and so on for the remaining off-diagonal covariance elements.

Instead of fixing a subset of factor loadings = 1, the factors can be scaled by fixing factor variances = 1, in which case the inter-factor covariance matrix is

$$\mathbf{\Psi} = \begin{bmatrix} 1 & & & \\ \sigma_{\eta_2\eta_1} & 1 & & \\ \sigma_{\eta_3\eta_1} & \sigma_{\eta_3\eta_2} & 1 & \\ \sigma_{\eta_4\eta_1} & \sigma_{\eta_4\eta_2} & \sigma_{\eta_4\eta_3} & 1 \end{bmatrix}$$

and the off-diagonal covariance elements are now in the correlation metric ranging between -1 and +1.

Finally, assuming the error terms are all uncorrelated, the error covariance matrix is

$$\mathbf{\Theta}_{\varepsilon} =$$

$$\begin{bmatrix} \sigma_{\varepsilon_1}^2 & \\ 0 & \sigma_{\varepsilon_2}^2 & & & & & & & & & & & & & & & & & & & \\ 0 & 0 & \sigma_{\varepsilon_3}^2 & & & & & & & & & & & & & & & & & & \\ 0 & 0 & 0 & \sigma_{\varepsilon_4}^2 & & & & & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & \sigma_{\varepsilon_5}^2 & & & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_6}^2 & & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_7}^2 & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_8}^2 & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_9}^2 & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{10}}^2 & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{11}}^2 & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{12}}^2 & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{13}}^2 & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{14}}^2 & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{15}}^2 & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{16}}^2 & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{17}}^2 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{18}}^2 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{19}}^2 \end{bmatrix}$$

where $\sigma_{\varepsilon_1}^2$ is the variance of ε_1 , that is, the variance of the error term in the equation for observed variable $Y_1 = TE1$,

$\sigma_{\varepsilon_2}^2$ is the error variance of $Y_2 = TE2$, and so on.

Matrix expression of one-factor model for PCS items

$$\begin{bmatrix} TE1 \\ TE2 \\ TE3 \\ TE4 \\ TE5 \\ OE1 \\ OE2 \\ OE3 \\ OE4 \\ LV1 \\ LV2 \\ LV3 \\ LV4 \\ EM1 \\ EM2 \\ EM3 \\ EM4 \\ EM5 \\ EM6 \end{bmatrix} = \begin{bmatrix} \lambda_{1,1} \\ \lambda_{2,1} \\ \lambda_{3,1} \\ \lambda_{4,1} \\ \lambda_{5,1} \\ \lambda_{6,1} \\ \lambda_{7,1} \\ \lambda_{8,1} \\ \lambda_{9,1} \\ \lambda_{10,1} \\ \lambda_{11,1} \\ \lambda_{12,1} \\ \lambda_{13,1} \\ \lambda_{14,1} \\ \lambda_{15,1} \\ \lambda_{16,1} \\ \lambda_{17,1} \\ \lambda_{18,1} \\ \lambda_{19,1} \end{bmatrix} [PC] + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \\ \varepsilon_9 \\ \varepsilon_{10} \\ \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{15} \\ \varepsilon_{16} \\ \varepsilon_{17} \\ \varepsilon_{18} \\ \varepsilon_{19} \end{bmatrix}$$

So the factor loading matrix is just a single column:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_{1,1} \\ \lambda_{2,1} \\ \lambda_{3,1} \\ \lambda_{4,1} \\ \lambda_{5,1} \\ \lambda_{6,1} \\ \lambda_{7,1} \\ \lambda_{8,1} \\ \lambda_{9,1} \\ \lambda_{10,1} \\ \lambda_{11,1} \\ \lambda_{12,1} \\ \lambda_{13,1} \\ \lambda_{14,1} \\ \lambda_{15,1} \\ \lambda_{16,1} \\ \lambda_{17,1} \\ \lambda_{18,1} \\ \lambda_{19,1} \end{bmatrix}$$

Because there is only one factor, the inter-factor covariance matrix consists of only a single element, which is the factor variance:

$$\mathbf{\Psi} = [\sigma_{\eta_1}^2]$$

The error covariance matrix is again a 19×19 diagonal matrix (assuming uncorrelated errors), like with the four-factor model.

Correlated error terms

In each model presented in Chapter 9, the error terms are uncorrelated, and therefore the $\mathbf{\Theta}_\epsilon$ matrix is diagonal, as shown above; i.e., all error covariances are fixed to 0 *a priori*. Often, modification indices (MIs) indicate that model fit will improve substantially if certain error covariance parameters are freed, allowing them to be nonzero. The data-driven practice of freeing error covariances (or any other parameter, for that matter) based on MIs is problematic because of a tendency to capitalize on sample-specific features of the data that can't be replicated when the same model with freed error covariances is fitted to a different sample. Nevertheless, freed error covariances are justifiable when they can be defended on substantive grounds and replicated with another sample.

To demonstrate freed error covariance parameters, we can revise the one-factor model presented above to allow free error covariances among the four outside effort items; indeed, the main text reports that the MI for the error covariance between *OE2* and *OE4* is a whopping 41.58 for the one-factor model.

The one-factor model scalar equation for observed variable $Y_7 = OE2$ is:

$$OE2_i = \lambda_{71}PC_i + \varepsilon_{7i}.$$

And the scalar equation for $Y_9 = OE4$ is:

$$OE4_i = \lambda_{91}PC_i + \varepsilon_{9i}.$$

We want to revise the one-factor model so that the error terms from these two equations, ε_7 and ε_9 , are freely correlated; that is, the covariance between ε_7 and ε_9 , is a free parameter rather than being fixed to 0. This error covariance corresponds to the element in the 9th row and 7th column of the Θ_ε error covariance matrix (and its mirror image in the 7th row and 9th column, because covariance matrices are symmetrical). Likewise, we might as well also free all error covariances among the outside effort items, that is, also free the covariances between ε_6 and ε_7 , between ε_6 and ε_8 , between ε_6 and ε_9 , between ε_7 and ε_8 , and so on. Then, the error covariance matrix becomes

$$\Theta_\varepsilon =$$

$$\begin{bmatrix} \sigma_{\varepsilon_1}^2 & & & & & & & & & & & & & & & & & & & \\ 0 & \sigma_{\varepsilon_2}^2 & & & & & & & & & & & & & & & & & & \\ 0 & 0 & \sigma_{\varepsilon_3}^2 & & & & & & & & & & & & & & & & & \\ 0 & 0 & 0 & \sigma_{\varepsilon_4}^2 & & & & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & \sigma_{\varepsilon_5}^2 & & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_6}^2 & & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_6, \varepsilon_7} & \sigma_{\varepsilon_7}^2 & & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_6, \varepsilon_8} & \sigma_{\varepsilon_7, \varepsilon_8} & \sigma_{\varepsilon_8}^2 & & & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_6, \varepsilon_9} & \sigma_{\varepsilon_7, \varepsilon_9} & \sigma_{\varepsilon_8, \varepsilon_9} & \sigma_{\varepsilon_9}^2 & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{10}}^2 & & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{11}}^2 & & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{12}}^2 & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{13}}^2 & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{14}}^2 & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{15}}^2 & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{16}}^2 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{17}}^2 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{18}}^2 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\varepsilon_{19}}^2 \end{bmatrix}$$

The online code demonstration file for Chapter 9 shows how to use R to estimate a revised one-factor model with these free error covariances. Results show improved model fit relative to the original one-factor model, with $SRMR = .06$, $CFI_{robust} = .88$, $TLI_{robust} = .86$, and $RMSEA_{robust} = .12$, although fit is still inadequate overall. The completely standardized error covariance estimates correspond to correlations; the estimated error correlations among the outside effort items are substantial, ranging from .40 to .58. But this model was estimated only for the purpose of pedagogical demonstration; the four-factor model (or the higher-order model presented below) provides a better representation of the factor structure for the 19 PCS item response variables.

Higher-order model for PCS items

As briefly described in the text, the PCS was developed to measure a broad construct, *psychological cost* while different subsets of items are meant to represent four specific aspects of cost. Specifically, one might hypothesize that a single *psychological cost* (*PC*) construct causes individual variability in four distinct, more specific domains or aspects of *PC*, namely *task effort*, *outside effort*, *loss of valued alternatives*, and *emotional cost*. In other words, a **higher-order factor** ($\xi = PC$) determines (unobservable) scores on three distinct **lower-order factors** ($\eta_1 = TE$, $\eta_2 = OE$, $\eta_3 = LV$, $\eta_4 = EM$), as depicted in Figure S9.1.

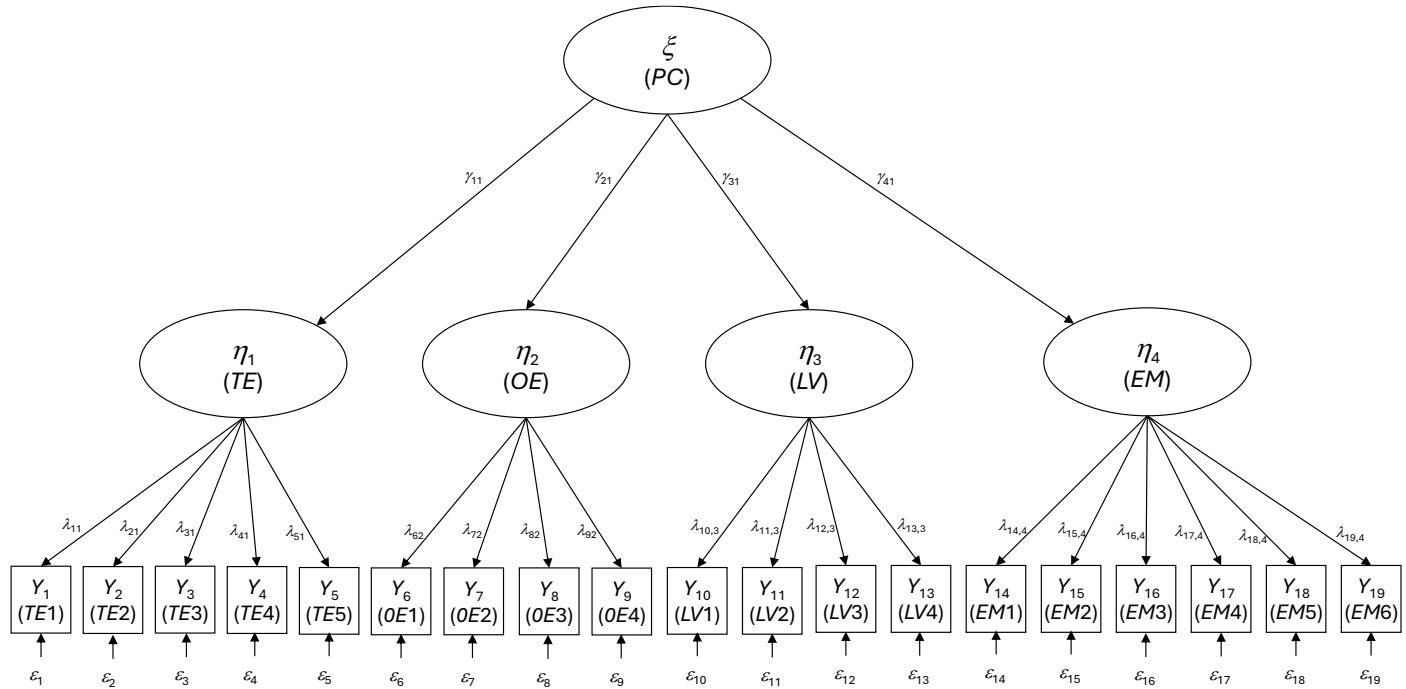


Figure S9.1. Higher-order model for the 19 PCS item response variables.

This model stipulates that each of the three lower-order factors (a.k.a. first-order factors) is linearly regressed on the higher-order factor (a.k.a. second-order factor) according to the equations below:

$$\eta_1 = \gamma_{11}\xi + \zeta_1$$

$$\eta_2 = \gamma_{21}\xi + \zeta_2$$

$$\eta_3 = \gamma_{31}\xi + \zeta_3$$

$$\eta_4 = \gamma_{41}\xi + \zeta_4$$

where the γ parameters are referred to as **higher-order factor loadings**.

The equations regressing each lower-order factor on the higher-order factor can be collected into a single matrix equation:

$$\boldsymbol{\eta} = \boldsymbol{\Gamma}\boldsymbol{\xi} + \boldsymbol{\zeta}$$

where $\boldsymbol{\eta}$ is the vector of lower-order factors, $\boldsymbol{\Gamma}$ is the higher-order factor loading matrix, $\boldsymbol{\xi}$ is the vector of higher-order factors, and $\boldsymbol{\zeta}$ is the vector of higher-order error terms. In the current example, and most applications, there is only one higher-order factor, so $\boldsymbol{\xi} = [\xi] = [PC]$, but it is possible to specify models with more than one higher-order factor provided that there are enough lower-order factors to ensure model identification (see Rindskopf & Rose, 1988). For the PCS example, this matrix equation is

$$\begin{bmatrix} TE \\ OE \\ LV \\ EM \end{bmatrix} = \begin{bmatrix} \gamma_{11} \\ \gamma_{21} \\ \gamma_{31} \\ \gamma_{41} \end{bmatrix} [\xi] + \begin{bmatrix} \zeta_1 \\ \zeta_2 \\ \zeta_3 \\ \zeta_4 \end{bmatrix}$$

The model also stipulates that the observed variables are directly regressed on the lower-order factors according to the matrix equation

$$\mathbf{Y} = \mathbf{\Lambda}\boldsymbol{\eta} + \boldsymbol{\varepsilon}$$

Thus, the structure of the lower-order factor loading matrix $\mathbf{\Lambda}$ for the higher-order model is the same as for the previous four-factor model (i.e., a 19×4 matrix), as is the error covariance matrix $\boldsymbol{\Theta}_{\boldsymbol{\varepsilon}}$.

A critical idea behind the higher-order model is that the higher-order factor accounts for the correlations among the lower-order factors. Accordingly, $\boldsymbol{\Psi}$ is a diagonal matrix such that

$$\boldsymbol{\Psi} = \begin{bmatrix} \sigma_{\zeta_1}^2 & & & \\ 0 & \sigma_{\zeta_2}^2 & & \\ 0 & 0 & \sigma_{\zeta_3}^2 & \\ 0 & 0 & 0 & \sigma_{\zeta_4}^2 \end{bmatrix},$$

specifying that the error variances of the lower-order factors are free parameters, but the covariances among errors of the lower-order factors are fixed to 0.

Taken together, the equations for the higher-order model indicate that the effect of the higher-order factor on a given observed variables is *mediated* by a lower-order factor; that is, the higher-order factor has a direct causal effect on each lower-order factor, and then the lower-order factors have direct effects on the observed variables.

Model identification is a complex issue with higher-order models. A rule-of-thumb is that there should be at least four lower-order factors associated with each higher-order factor; in the current example, there are indeed four lower-order factors caused by the *PC* higher-order factor. A model with one higher-order factor and only three lower-order factors is statistically equivalent to a model with three correlated factors, whereas a model with one higher-order factor and only two lower-order factors is under-identified. See Rindskopf and Rose (1988) for further details.

The online code demonstration file for Chapter 9 shows how to estimate the higher-order model in R. Like any CFA model, it is necessary to include parameter restrictions to set the factors' scales. Applying the marker-variable approach, each lower-order factor has a factor loading fixed to equal 1, and we also set one of the higher-order factor loadings equal to 1; for the PCS example, we can fix $\gamma_{11} = 1$.

When this higher-order model is fitted to the PCS data, $SRMR = .05$, $CFI_{robust} = .97$, $TLI_{robust} = .97$, and $RMSEA_{robust} = .06$, all of which indicate good model fit. The Satorra-Bentler scaled χ^2 difference test comparing the higher-order model with the original four-factor model presented in the main text is $SB \chi^2_{difference}(2) = 5.44$, $p = .07$, indicating that the more parsimonious higher-order model does not fit the data significantly worse than the four-factor model. Furthermore, the BIC statistic is slightly lower for the higher-order model ($BIC = 7157.3$) than for the four-factor model ($BIC = 7159.1$), although AIC favours the four-factor model ($AIC = 7025.5$) over the higher-order model ($AIC = 7029.7$).

Table S9.1 gives the factor-loading estimates for the higher-order model fitted to the PCS item-level data. Notice that the marker-variable approach was used to scale the factors, but as mentioned in Chapter 9, the completely standardized estimates are the same regardless of how the factors' scales are set. Of note is that the (standardized) higher-order factor loadings are all strong, ranging from 0.77 to 0.95, indicating that each lower-order factor is strongly influenced by the higher-order factor; specifically, a broad *psychological cost* construct strongly determines each of the lower-order factors representing narrower aspects of psychological cost.

Table S9.1

Factor-loading estimates from the higher-order model of PCS items

Higher-order factor loadings								
	<i>TE</i> (η_1)		<i>OE</i> (η_2)		<i>LV</i> (η_3)		<i>EM</i> (η_4)	
	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$
<i>PC</i> (ξ)	1	0.95	0.76	0.77	0.83	0.89	0.93	0.89
Lower-order factor loadings								
Observed <i>Y</i>	<i>TE</i> (η_1)		<i>OE</i> (η_2)		<i>LV</i> (η_3)		<i>EM</i> (η_4)	
	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$
<i>TE1</i>	1	0.89	0	0	0	0	0	0
<i>TE2</i>	1.02	0.85	0	0	0	0	0	0
<i>TE3</i>	0.84	0.79	0	0	0	0	0	0
<i>TE4</i>	0.91	0.85	0	0	0	0	0	0
<i>TE5</i>	0.95	0.85	0	0	0	0	0	0
<i>OE1</i>	0	0	1	0.81	0	0	0	0
<i>OE2</i>	0	0	0.96	0.83	0	0	0	0
<i>OE3</i>	0	0	0.98	0.82	0	0	0	0
<i>OE4</i>	0	0	1.01	0.88	0	0	0	0
<i>LV1</i>	0	0	0	0	1	0.82	0	0
<i>LV2</i>	0	0	0	0	1.03	0.93	0	0
<i>LV3</i>	0	0	0	0	0.96	0.83	0	0
<i>LV4</i>	0	0	0	0	1.04	0.81	0	0
<i>EM1</i>	0	0	0	0	0	0	1	0.81
<i>EM2</i>	0	0	0	0	0	0	0.91	0.85
<i>EM3</i>	0	0	0	0	0	0	1.02	0.85
<i>EM4</i>	0	0	0	0	0	0	0.95	0.86
<i>EM5</i>	0	0	0	0	0	0	1.05	0.90
<i>EM6</i>	0	0	0	0	0	0	1.17	0.86

Note. $\hat{\lambda}$ = unstandardized factor-loading estimate. $\hat{\lambda}^*$ = completely standardized factor-loading estimate. 1 indicates parameters fixed to 1 *a priori*; 0 indicates parameters fixed to 0 *a priori*.

Bifactor model for PCS items

Bifactor models are very popular in scale development and validation studies. A **bifactor model** may be considered an extension of a one-factor model in that there is a single, dominant factor which directly influences *all* observed variables; this factor is known as a **general factor**. Yet, the general factor may not be sufficient to explain all of the covariances among certain clusters of observed variables. Thus, **specific factors** (also sometimes called **group factors**) are specified to explain the covariances among observed variables which are not accounted for by the general factor.

As an alternative to a higher-order model, one could hypothesize that responses to all 19 PCS items are directly caused by a single, general *psychological cost* construct, but there is excess, shared covariance among items from the same domain of cost. For example, all five *task effort* items are directly influenced by a general psychological cost factor, but because they are based

on the same domain of cost, there is excess covariance among the task effort items that is independent of the general cost factor; a specific task effort factor influences the five task effort items above and beyond the general cost factor. Likewise, the four *outside effort* item responses are also influenced by the general psychological cost factor, but there is also a specific outside effort factor which explains excess covariance among the outside effort items that is separate from the general cost factor. Then there are analogous specific factors explaining excess covariance among the loss of valued alternatives items and the emotional effort items. This hypothesized bifactor structure is depicted in Figure S9.2.

It's important to keep in mind that the four specific factors do not have the same interpretation as the four correlated factors in the basic four-factor model presented in the main text of Chapter 9, nor do they have the same interpretation as the lower-order factors in the higher-order model. For instance, in the basic four-factor model and the higher-order model, $\eta_1 = TE$ represents the task effort construct, conceptualized as an aspect of a broader psychological cost construct by Flake et al. (2015), but in the bifactor model, η_1 represents the correlated portions of the task effort items which are *independent* from the general psychological cost construct. Hence, it is critical that the specific factors be uncorrelated with the general factor, as shown in the Ψ matrix below.

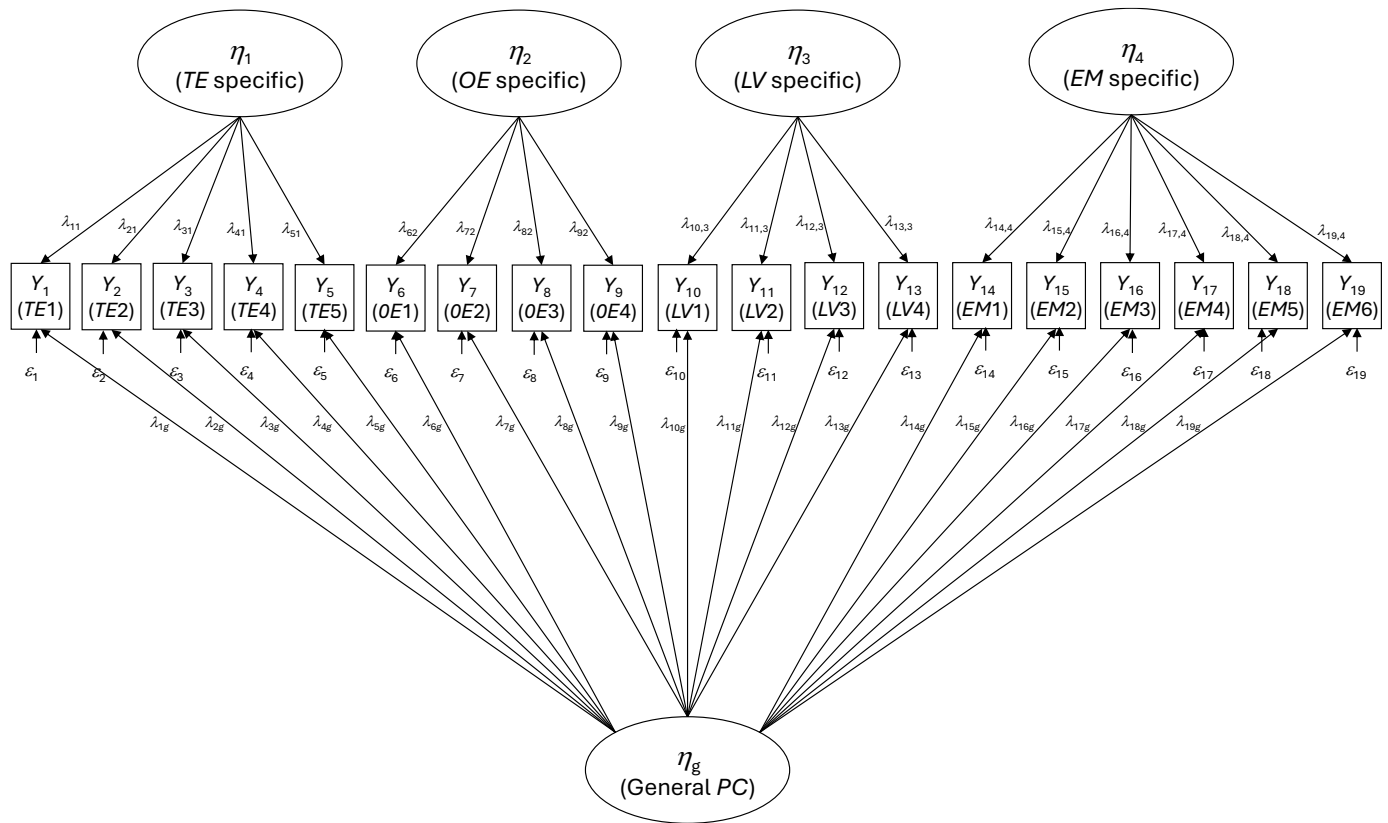


Figure S9.2

The name *bifactor* makes sense because each observed variable is directly influenced by two latent variables – the general factor and one of the specific factors – despite that the complete model has more than two factors. Thus, a bifactor model stipulates that each observed variable is linearly regressed on both the general factor and a specific factor. The generic CFA model

equations (i.e., Equations 9.1 and 9.2 in the main text) are sufficient to specify a bifactor model given the right pattern of fixed and free parameters. Using the example bifactor model for the PCS items in Figure S9.2, the first item, $Y_1 = TE1$, is hypothesized to be influenced by both the general factor and a specific factor capturing error covariances among the task effort items. Thus, applying Equation 9.1, the scalar regression equation for Y_1 is

$$Y_1 = \lambda_{1g}\eta_g + \lambda_{11}\eta_1 + 0\eta_2 + 0\eta_3 + 0\eta_4 + \varepsilon_1$$

where η_g is the general factor and η_1 is a specific factor representing the covariance among task effort items that is not explained by the general factor. Of course, this equation can be simplified to

$$Y_1 = \lambda_{1g}\eta_g + \lambda_{11}\eta_1 + \varepsilon_1.$$

In comparison, $Y_6 = OE1$ is hypothesized to be influenced by the general factor and a specific factor capturing error covariances among the outside effort items. Thus, the scalar regression equation for Y_6 is

$$Y_6 = \lambda_{6g}\eta_g + 0\eta_1 + \lambda_{62}\eta_2 + 0\eta_3 + 0\eta_4 + \varepsilon_6,$$

or simply

$$Y_6 = \lambda_{6g}\eta_g + \lambda_{62}\eta_2 + \varepsilon_6.$$

where η_g is again the general factor and η_2 is a specific factor for the outside effort items.

The complete matrix regression equation for the PCS bifactor model is

$$\begin{bmatrix} TE1 \\ TE2 \\ TE3 \\ TE4 \\ TE5 \\ OE1 \\ OE2 \\ OE3 \\ OE4 \\ LV1 \\ LV2 \\ LV3 \\ LV4 \\ EM1 \\ EM2 \\ EM3 \\ EM4 \\ EM5 \\ EM6 \end{bmatrix} = \begin{bmatrix} \lambda_{1g} & \lambda_{1,1} & 0 & 0 & 0 \\ \lambda_{2g} & \lambda_{2,1} & 0 & 0 & 0 \\ \lambda_{3g} & \lambda_{3,1} & 0 & 0 & 0 \\ \lambda_{4g} & \lambda_{4,1} & 0 & 0 & 0 \\ \lambda_{5g} & \lambda_{5,1} & 0 & 0 & 0 \\ \lambda_{6g} & 0 & \lambda_{6,2} & 0 & 0 \\ \lambda_{7g} & 0 & \lambda_{7,2} & 0 & 0 \\ \lambda_{8g} & 0 & \lambda_{8,2} & 0 & 0 \\ \lambda_{9g} & 0 & \lambda_{9,2} & 0 & 0 \\ \lambda_{10g} & 0 & 0 & \lambda_{10,3} & 0 \\ \lambda_{11g} & 0 & 0 & \lambda_{11,3} & 0 \\ \lambda_{12g} & 0 & 0 & \lambda_{12,3} & 0 \\ \lambda_{13g} & 0 & 0 & \lambda_{13,3} & 0 \\ \lambda_{14g} & 0 & 0 & 0 & \lambda_{14,4} \\ \lambda_{15g} & 0 & 0 & 0 & \lambda_{15,4} \\ \lambda_{16g} & 0 & 0 & 0 & \lambda_{16,4} \\ \lambda_{17g} & 0 & 0 & 0 & \lambda_{17,4} \\ \lambda_{18g} & 0 & 0 & 0 & \lambda_{18,4} \\ \lambda_{19g} & 0 & 0 & 0 & \lambda_{19,4} \end{bmatrix} \begin{bmatrix} \eta_g \\ TE \\ OE \\ LV \\ EM \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \\ \varepsilon_9 \\ \varepsilon_{10} \\ \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{15} \\ \varepsilon_{16} \\ \varepsilon_{17} \\ \varepsilon_{18} \\ \varepsilon_{19} \end{bmatrix}.$$

Importantly, the covariances between the general factor and the specific factors must be fixed to zero for model identification; additionally, although it is not necessary for the covariance

between the specific factors to be set to zero, the traditional bifactor model does so. Therefore, Ψ is fixed to be a diagonal matrix:

$$\Psi = \begin{bmatrix} \text{VAR}(\eta_g) & 0 & 0 & 0 & 0 \\ 0 & \text{VAR}(\eta_1) & 0 & 0 & 0 \\ 0 & 0 & \text{VAR}(\eta_2) & 0 & 0 \\ 0 & 0 & 0 & \text{VAR}(\eta_3) & 0 \\ 0 & 0 & 0 & 0 & \text{VAR}(\eta_4) \end{bmatrix}$$

Furthermore, the error covariance matrix Θ_{ϵ} is once again constrained to be diagonal.

Again, we also must set the latent variable scales for model identification. The marker-variable approach (i.e., fixing a factor loading = 1 for each column of the factor loading matrix) can be used or the factor variances (i.e., diagonal elements of Ψ) can be fixed to equal 1, among other possibilities.

The online code demonstration file for Chapter 9 shows how to estimate the bifactor model in R. When the bifactor model is fitted to the PCS data, $SRMR = .04$, $CFI_{robust} = .98$, $TLI_{robust} = .97$, and $RMSEA_{robust} = .05$, all of which indicate good model fit. Table S9.2 gives the factor-loading estimates for the bifactor model fitted to the PCS item-level data. Notice again that the marker-variable approach was used to scale the factors, but as mentioned in Chapter 9, the completely standardized estimates are the same regardless of how the factors' scales are set. The (standardized) general factor loadings are all strong, ranging from 0.61 to 0.84 and each item's general factor loading is stronger than its specific factor loading, suggesting that the 19 PCS items each contribute substantially to the measurement of a general psychological cost factor. Yet, many of the items also have moderately large specific factor loadings, suggesting that each item domain captures substantial shared covariance that's completely independent of the psychological cost factor.

Table S9.2

Factor-loading estimates from the bifactor model of PCS items

Observed Y	Factors									
	General PC		TE specific		OE specific		LV specific		EM specific	
	(η_g)		(η_1)		(η_2)		(η_3)		(η_4)	
	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$	$\hat{\lambda}$	$\hat{\lambda}^*$
$TE1$	1	0.84	1	0.29	0	0	0	0	0	0
$TE2$	1.00	0.79	1.28	0.34	0	0	0	0	0	0
$TE3$	0.81	0.72	1.15	0.34	0	0	0	0	0	0
$TE4$	0.95	0.83	0.46	0.14	0	0	0	0	0	0
$TE5$	0.96	0.82	0.76	0.22	0	0	0	0	0	0
$OE1$	0.76	0.63	0	0	1	0.50	0	0	0	0
$OE2$	0.79	0.70	0	0	0.82	0.43	0	0	0	0
$OE3$	0.71	0.61	0	0	1.07	0.55	0	0	0	0
$OE4$	0.71	0.64	0	0	1.18	0.63	0	0	0	0
$LV1$	0.90	0.80	0	0	0	0	1	0.22	0	0
$LV2$	0.83	0.80	0	0	0	0	2.27	0.53	0	0
$LV3$	0.78	0.72	0	0	0	0	1.67	0.37	0	0
$LV4$	0.71	0.70	0	0	0	0	2.09	0.42	0	0
$EM1$	0.93	0.71	0	0	0	0	0	0	1	0.37
$EM2$	0.89	0.80	0	0	0	0	0	0	0.69	0.30
$EM3$	0.92	0.73	0	0	0	0	0	0	1.12	0.43
$EM4$	0.85	0.73	0	0	0	0	0	0	1.11	0.46
$EM5$	1.00	0.82	0	0	0	0	0	0	0.95	0.38
$EM6$	1.07	0.75	0	0	0	0	0	0	1.29	0.44

Note. $\hat{\lambda}$ = unstandardized factor-loading estimate. $\hat{\lambda}^*$ = completely standardized factor-loading estimate. 1 indicates parameters fixed to 1 *a priori*; 0 indicates parameters fixed to 0 *a priori*.

Distinguishing higher-order models from bifactor models

The higher-order and bifactor models are similar in that both include a general, broad factor that influences each observed variable.

But in a higher-order model, the higher-order factor influences observed variables only *indirectly*: Each observed variable is directly influenced by only one lower-order factor (assuming an independent clusters solution), but then the lower order factors are influenced by the higher-order factor. In a bifactor model, each observed variable is *directly* influenced by both the general factor and a specific factor.

Because the specific factors represent the portion of explained variance over and above variance due to the general factor, the bifactor model is *not* consistent with a subscale conceptualization. That is, the specific factors do not represent the constructs that would be implied by subscale scores; rather, the lower-order factors of a higher-order model make sense as representations of constructs implied by subscales.

Often, the bifactor model is used to represent the “essential unidimensionality” of a set of observed variables (represented with the general factor) while including the specific factors to represent excess covariance (indeed, *error* covariance) among subsets of observed variables. With the PCS example, the general factor loadings are consistently strong across all items, providing some evidence of this essential unidimensionality concept, although a summary measure such as *omega-hierarchical* could be used to assess the overall reliability of the PCS as a measure of general cost factor that’s common to all 19 items (see Flora, 2020, for example).

There is, however, a formal mathematical association between higher order and bifactor models, as explained by Yung et al. (1999). In fact, the higher-order model for the PCS items is nested within the bifactor model, and therefore we can use a likelihood ratio test (i.e., chi-square difference test) to formally compare their fit to the data. In the PCS example, the Satorra-Bentler scaled χ^2 difference test comparing the higher-order model with the bifactor model is $SB \chi^2_{difference}(15) = 23.48, p = .07$, indicating that the more parsimonious higher-order model does not fit the data significantly worse than the bifactor model. Furthermore, the *BIC* statistic is lower for the higher-order model ($BIC = 7157.3$) than for the bifactor model ($BIC = 7200.7$), although *AIC* favours the bifactor model ($AIC = 7027.6$) over the higher-order model ($AIC = 7029.7$).

References

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